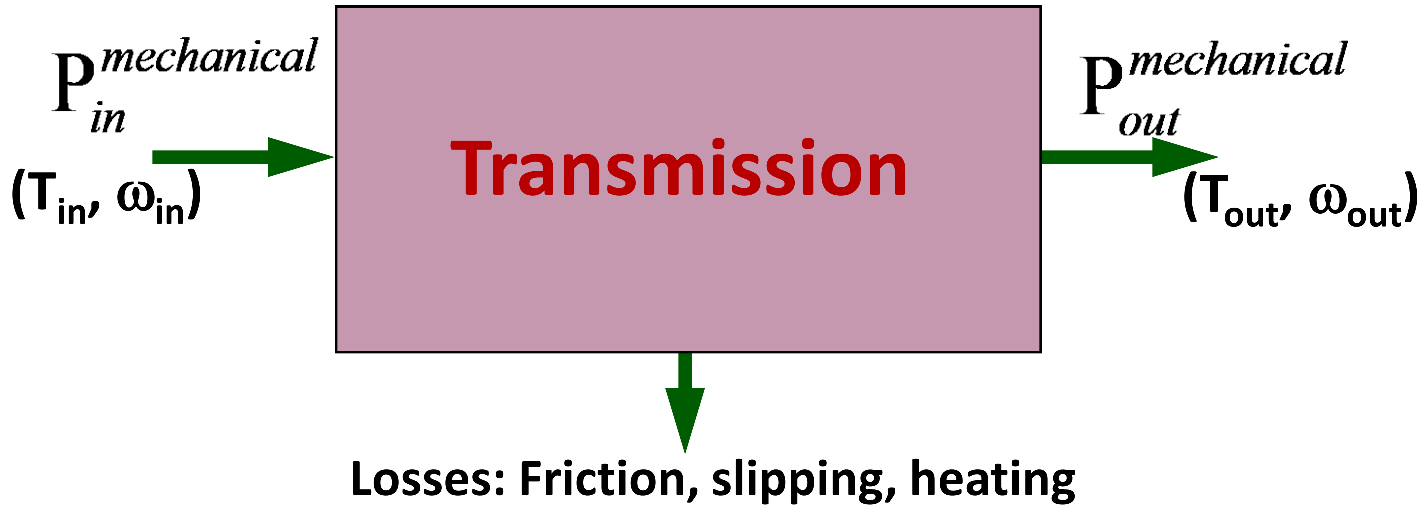
ME 350 Lecture #8 Introduction to Transmissions Gear Analysis

- Types of Transmissions
- Gear Terminology
- Gear Ratio
- Compound Gears
- Planetary Gears

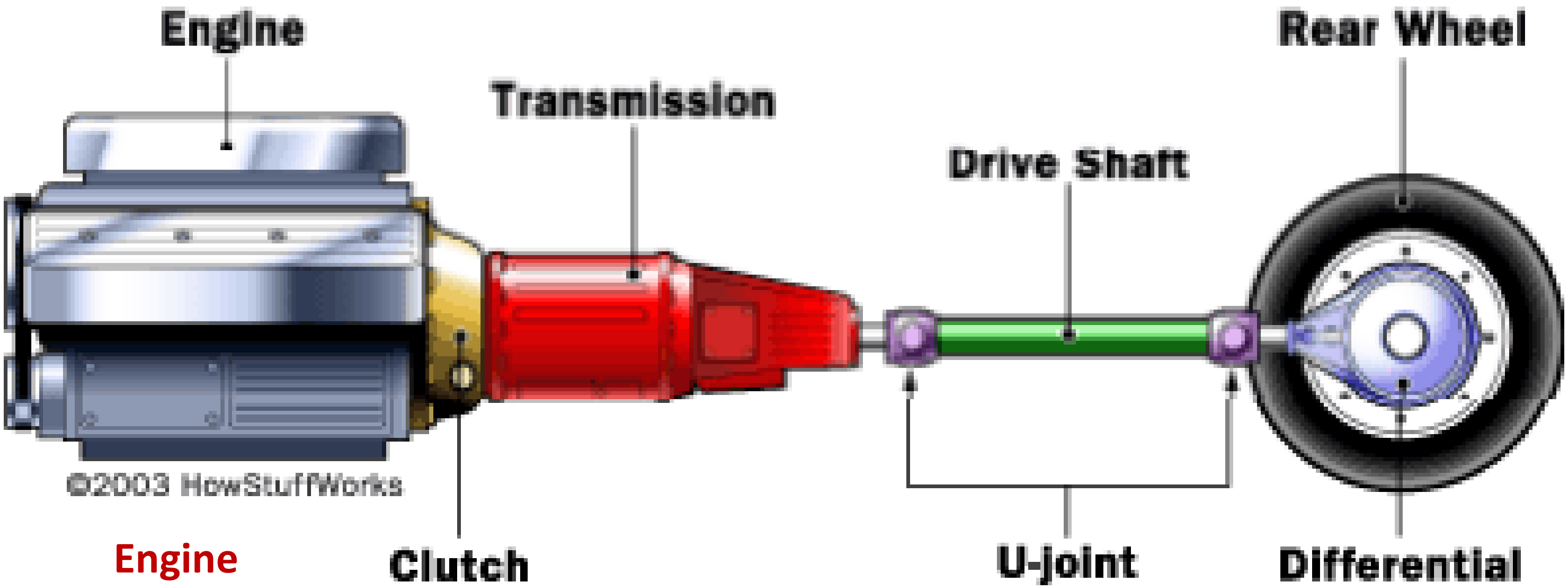
Prof. Diann Brei

What is a (mechanical) transmission?

A transmission is a device that **transfers** (mechanical) power from one point to the other.



Why are Transmissions Needed?



Engine
800-6000 rpm

Wheel
0-2000 rpm

Transmissions can be used to change:

- Force/torque
- Location/orientation of motion
- Form of motion (i.e., linear to rotational)
- Speed
- etc…

Break Exercise

- Get in groups of 2 to 4
- Go to the different “grave yards” and observe the different types of transmission elements
- Compare gears, belts and drives on how well they perform across different metrics.
- To get you started – think about torques, packaging, cost, sound, lifetime, efficiency – *ETC*.
- Be prepared to share some with the class

Examples of Transmissions: Gear Drives

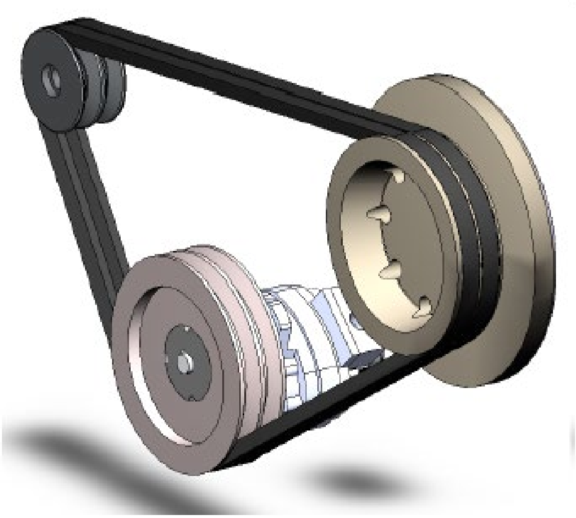
Gears change the rotational velocity and the torque to suit the motor and load conditions. Gears can also be used to simply transport power from one shaft to another shaft. Gears are highly efficient with the output power approximately equal to the input power.

- Can transmit large torques
- Speed ratio is constant
- Highly efficient
- Lower maintenance
- Relatively expensive
- May be noisy due to teeth meshing
- Transmits shocks directly (no shock isolation)

Examples of Transmissions: Belt Drives

Used where rotational speeds are relatively high. The linear speed of a belt is usually 2,500 to 7,000 ft/min with ideal around 4,000 ft/min. At lower speeds, the tension in the belt becomes too large for typical belt cross sections and at higher speeds, dynamic effects such as centrifugal forces, belt whip, and vibration reduce effectiveness of drive and its life.

- Provides flexibility to the designer (e.g., center distance)
- Reduced cost compared to gears
- Absorbs shock and vibration (isolates input from output)
- Quieter operation
- Speed ratio may vary due to slippage (except
- Shorter life span than gears
- Limited torque transmission due to slip
- Lower efficiency (depends on type)

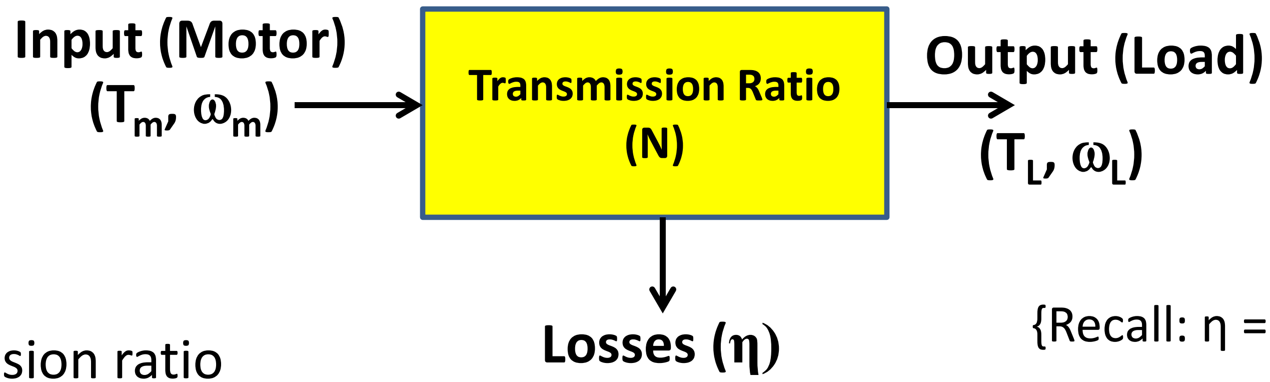


Examples of Transmissions: Chain Drives

Used at lower speed where high torques are needed. At higher speeds, noise, impact between chain link and sprocket teeth and lubrication become major problems.

- Design flexibility (e.g., center distance)
- Can transmit large torques
- Can maintain consistent speed ratio
- Could be noisy due to meshing
- Lower speeds due to inertia
- Lubrication needed

Transmission Ratio



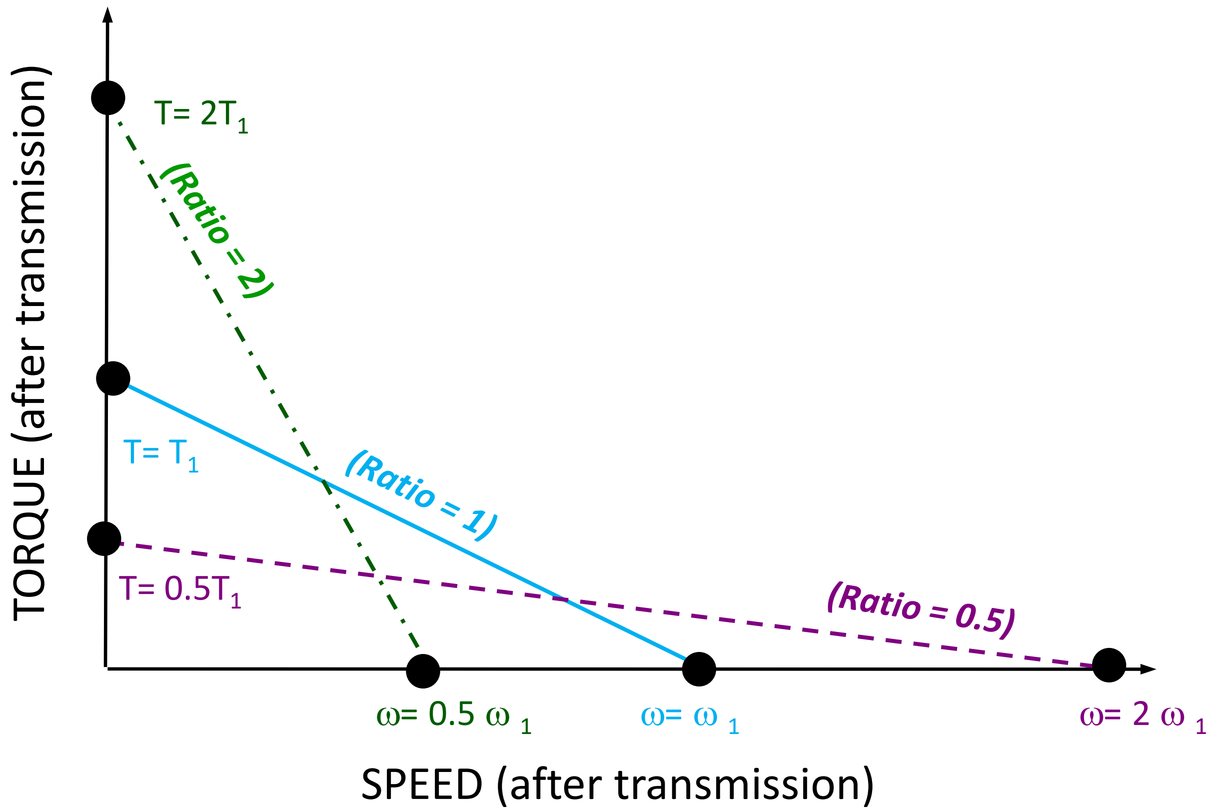
{Recall: $\eta = P_{out} / P_{in}$}

Transmission ratio

$$\omega_m / \omega_L = T_L / (\eta T_m) = T_L / T_m$$

Assuming 100% efficiency

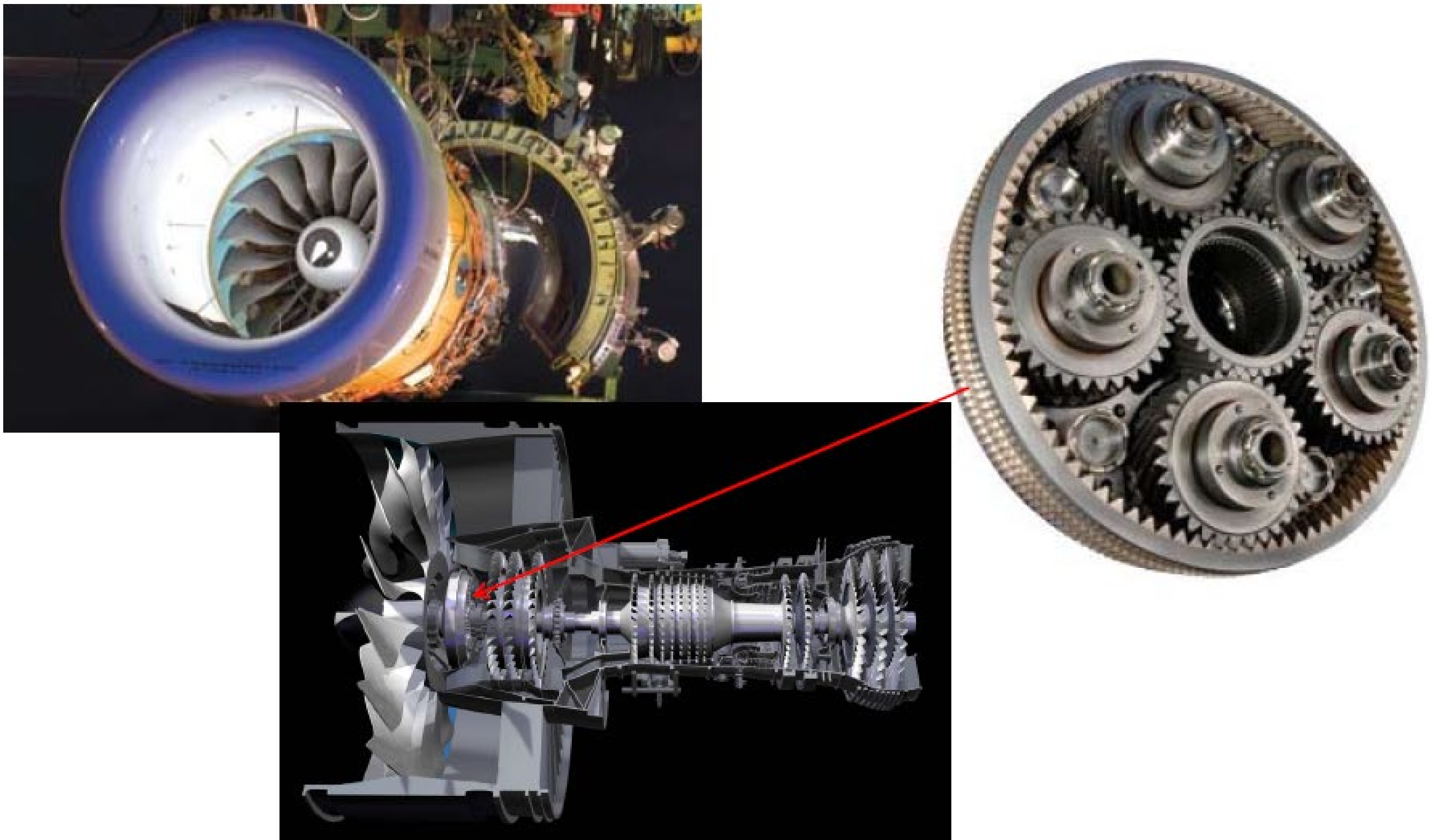
Effect of Transmission on Speed-Torque Curve



Transmission ratio can be used to change the **Speed-torque gradient** of a motor's speed-torque curve without changing **Motor windings**.
However, this could come at the cost of **Size and Efficiency**

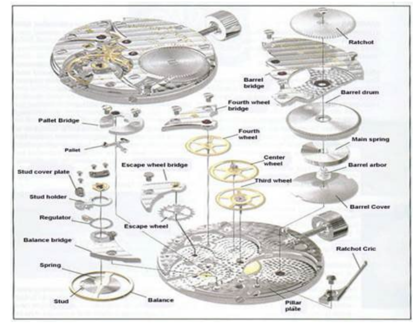
Glorious Gears

TurboFan Drive Speed Reduction (Gear)



Mechanical Watch

http://www.thepurists.com/watch/features/8ohms/jlchw/movement-007.jpg



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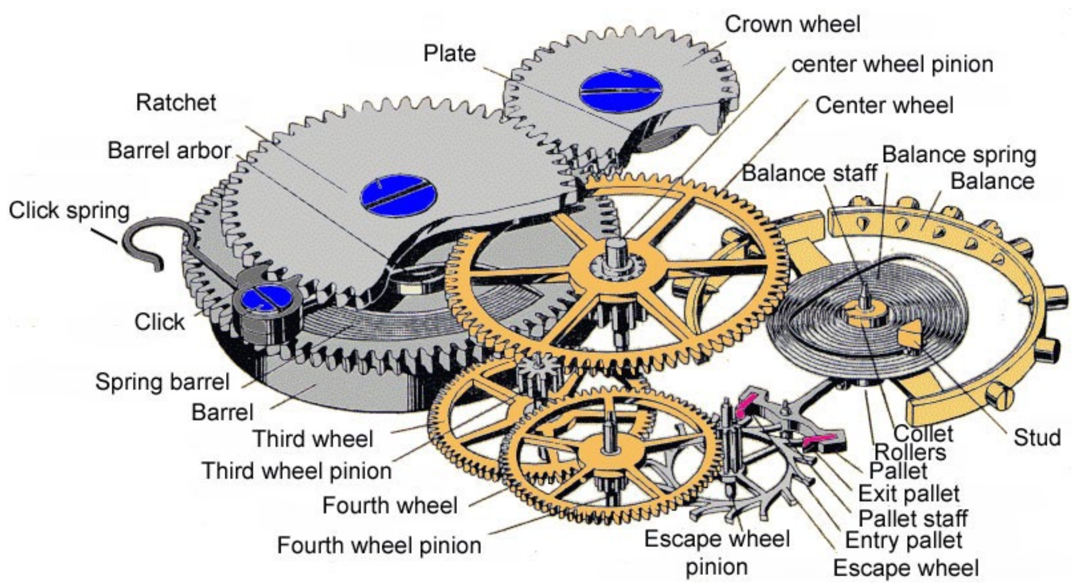
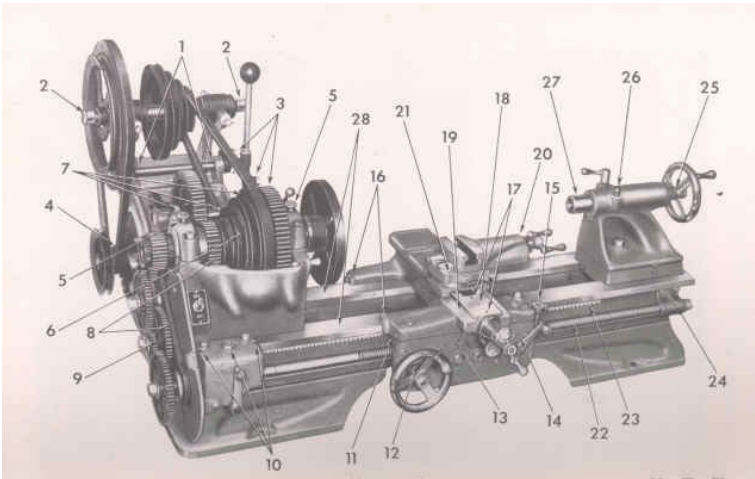
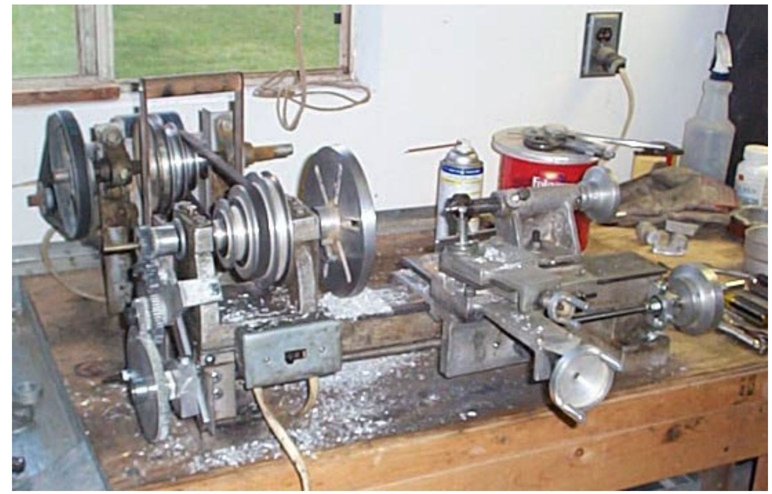


http://www.horologist.com/images/GearTrain02.jpg

Lathe



http://www.quarterbore.com/store/cd/atlaslathe/lathe_2.jpg



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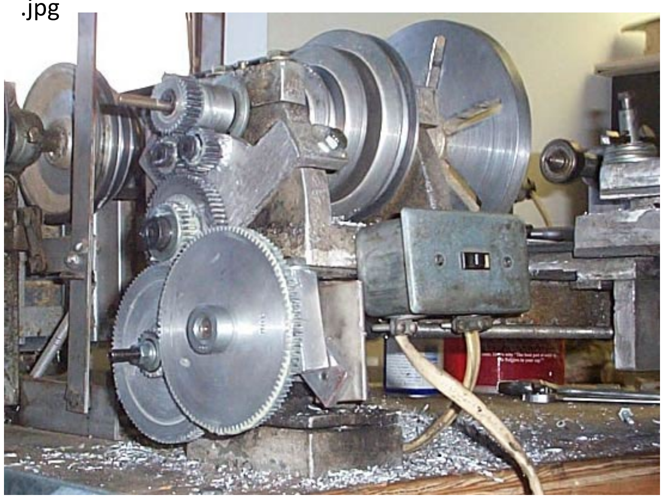
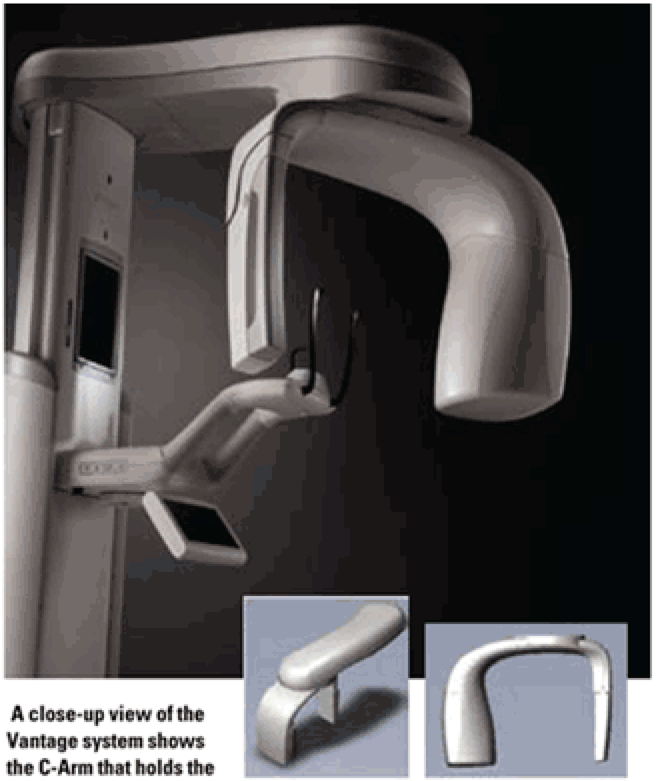


http://home.earthlink.net/~jschwytzer/side_oblique.jpg

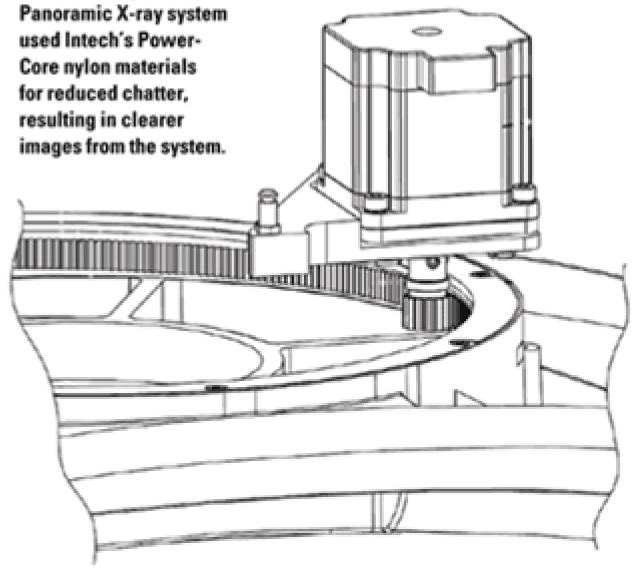
X-ray Machine



A close-up view of the Vantage system shows the C-Arm that holds the lasers as well as the removable CCD receptor.
(Inset) The overhead assembly and C-Arm are integral to the overall image quality of the Progeny Vantage Panoramic X-ray System, and incorporate specially designed Intech Power-Core components to increase accuracy.

http://medicaldesign.com/engineering-prototyping/research-development/0911-Imaging-vantage-system.gif

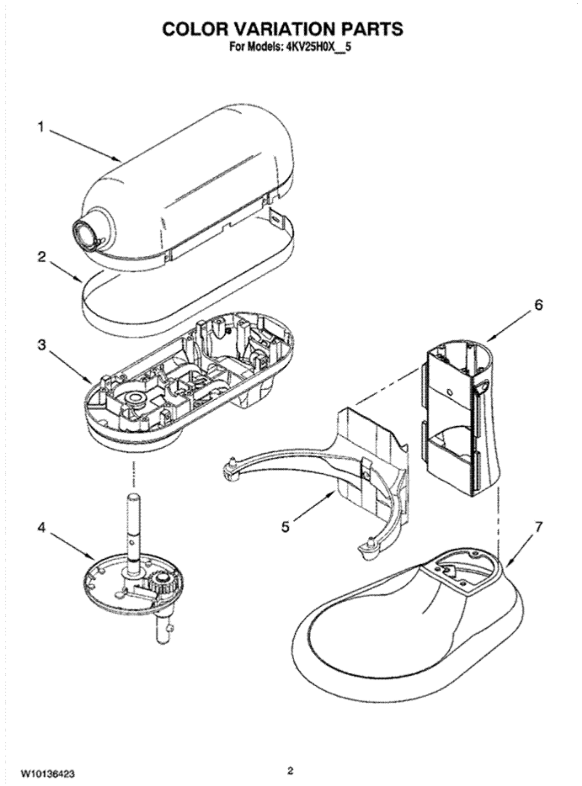
The precision gearing for the Vantage Panoramic X-ray system used Intech's Power-Core nylon materials for reduced chatter, resulting in clearer images from the system.



http://medicaldesign.com/engineering-prototyping/research-development/0911-Imaging-precision-gearing.gif

Stand Mixer

http://1.bp.blogspot.com/__nK5eqVEjqs/S2D8qrfo9yI/AAAAAAAAAts/tVOPnU6J-Js/s400/Artisan+Stand+Mixers.jpg



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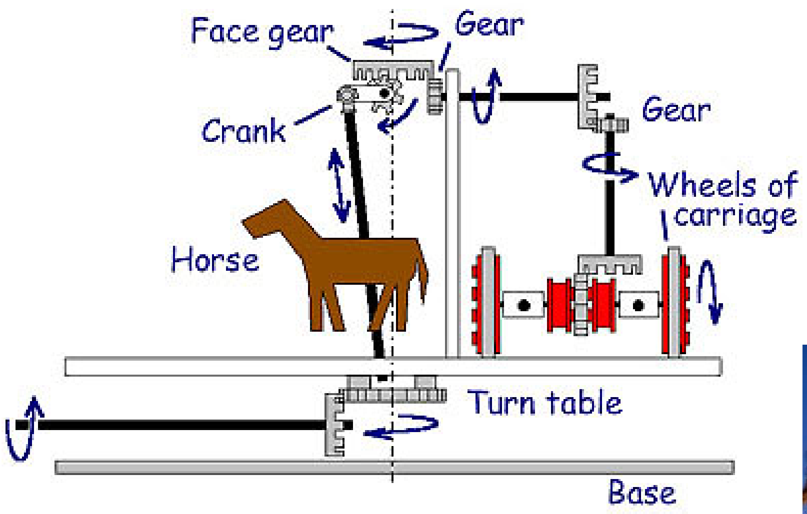
Marry Go Round

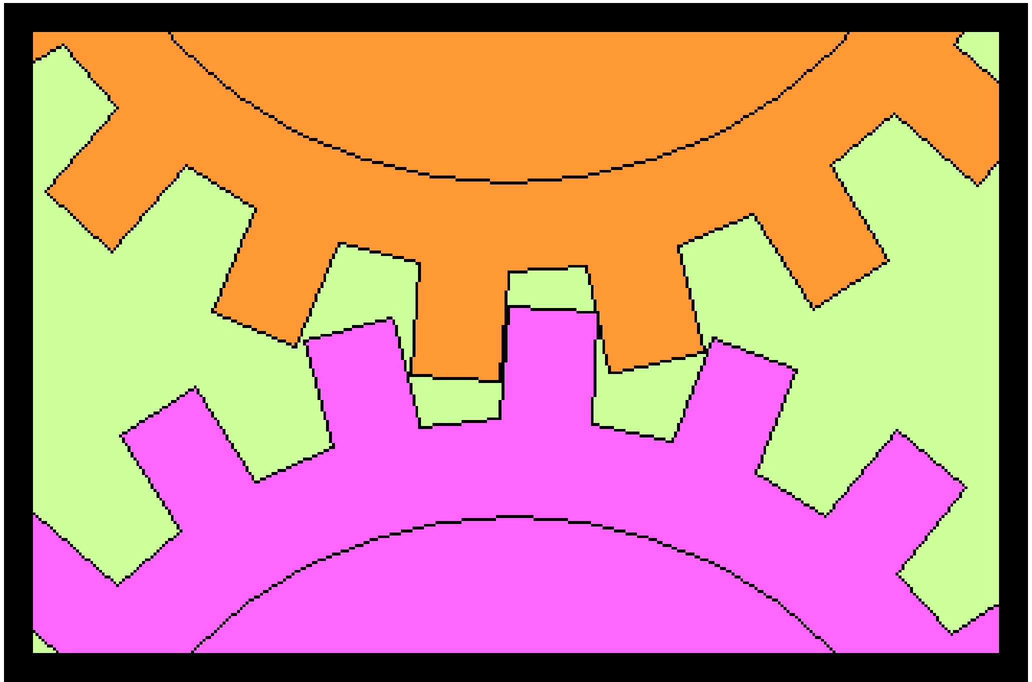


http://www21.ocn.ne.jp/~glass-cg/mry2_9.jpg

http://www.worldofstock.com/slides/TAE2542.jpg

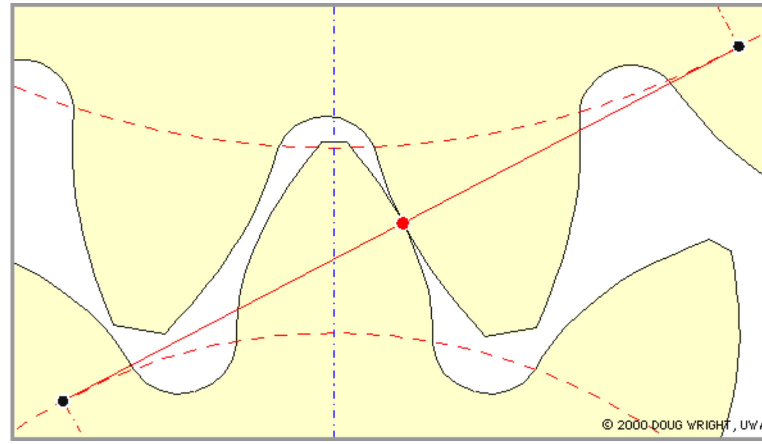
The Involute Profile

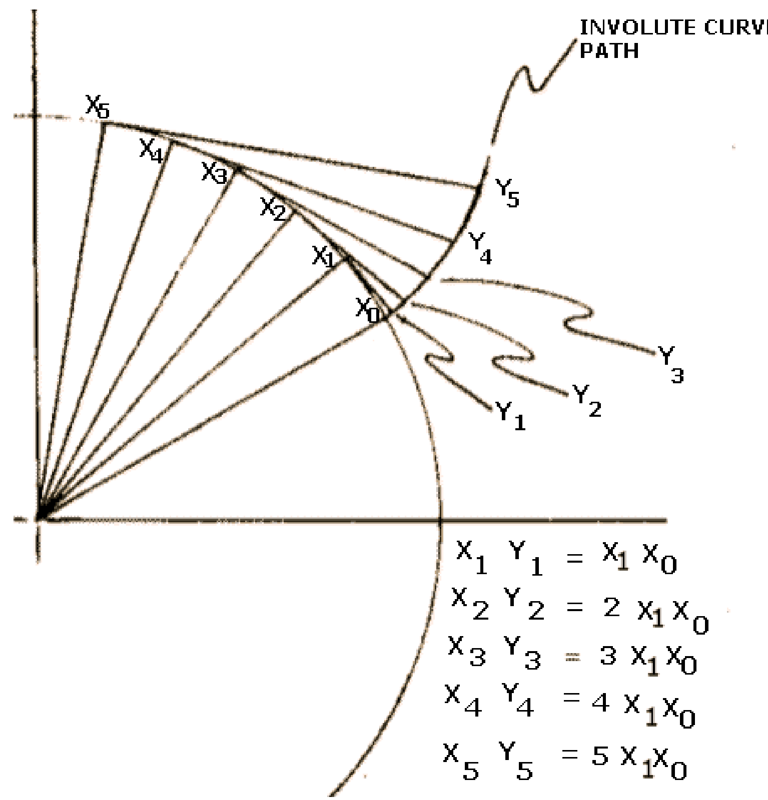
- If two meshing gear were manufactured with square teeth instead of being cut to an involute form, the gears would bind and not be able to rotate in mesh.
- Two such gears would lock together making assembly difficult, and rotation impossible. Note how the gears are locked together.
- If enough clearance were created to enable rotation, the speed of the output gear would be inconsistent, and backlash in the system would be an issue.



The Involute Profile

- An involute is constructed such that constant angular velocity (rotation rate) is maintained throughout gear contact.
- This constant angular velocity is what is known as ***conjugate action***
- An involute profile can be constructed by pinning a string to a wheel and then unraveling it as shown in the following figure.
- For reasons of economy in production, modern gear teeth are almost exclusively cut to an involute form.



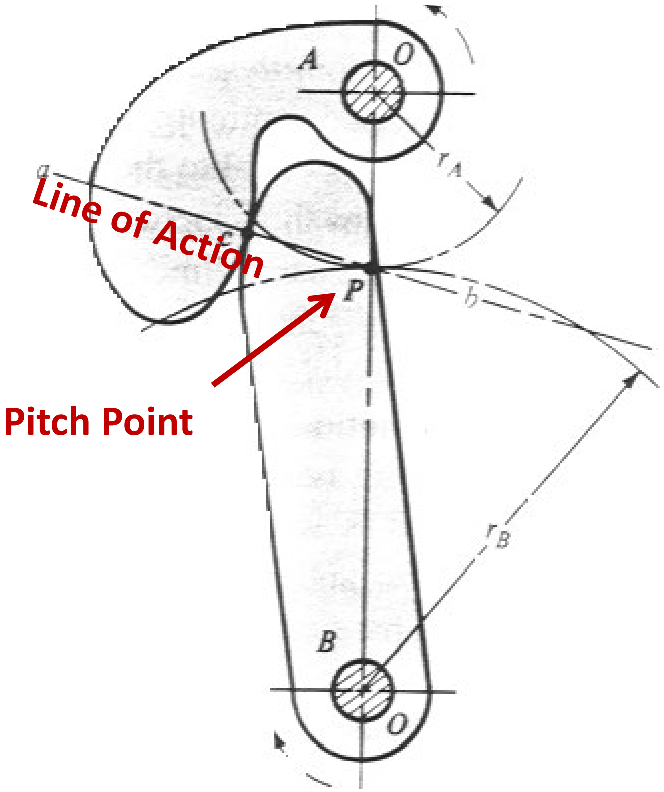


Conjugate Action

•It is important to have a constant velocity ratio; otherwise, there is severe vibration, impact and noise problems.

•When one curved surface pushes against another, the point of contact occurs where the two surfaces are tangent to each other and the forces at any instant are directed along the common normal *ab* to the two curves, called the line of action.

•The *line of action* intersects the *line of the centers* O-O at point P called the pitch point. The angular velocity ratio between the two arms are inversely proportional to their radii to the pitch point. Circles drawn through point P from each center are called the pitch circles and the radius of each circle is called the pitch radius.



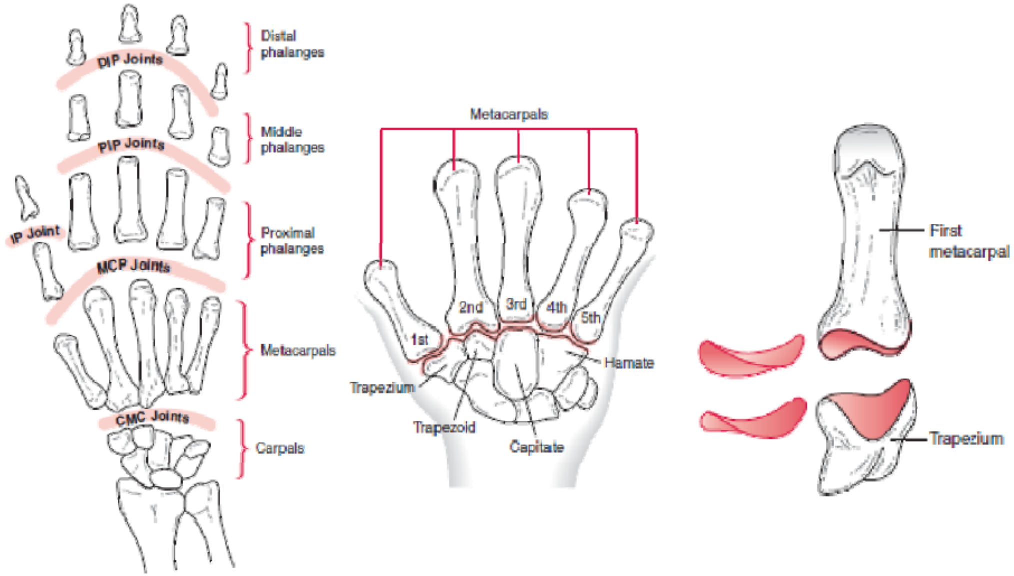
Law of Gearing

The law of gearing states that *if* the line drawn perpendicular to the surfaces of two rotating bodies at their point of contact *always* crosses the centerline between the two bodies at the same place, the angular velocity ratio of the two bodies will be constant.

Conjugate Action

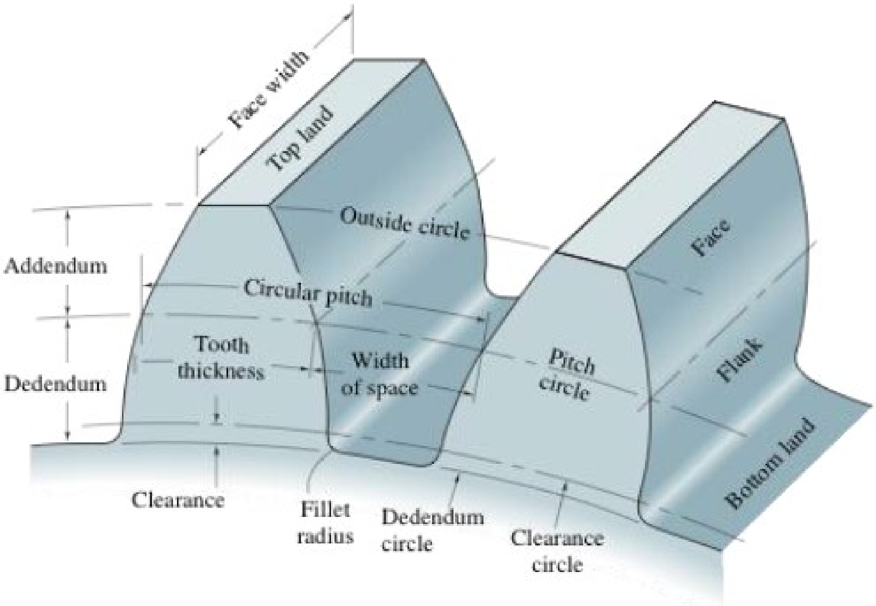
Other complex profiles have conjugate actions

- Carpometacarpal (CMC) joints of the thumb

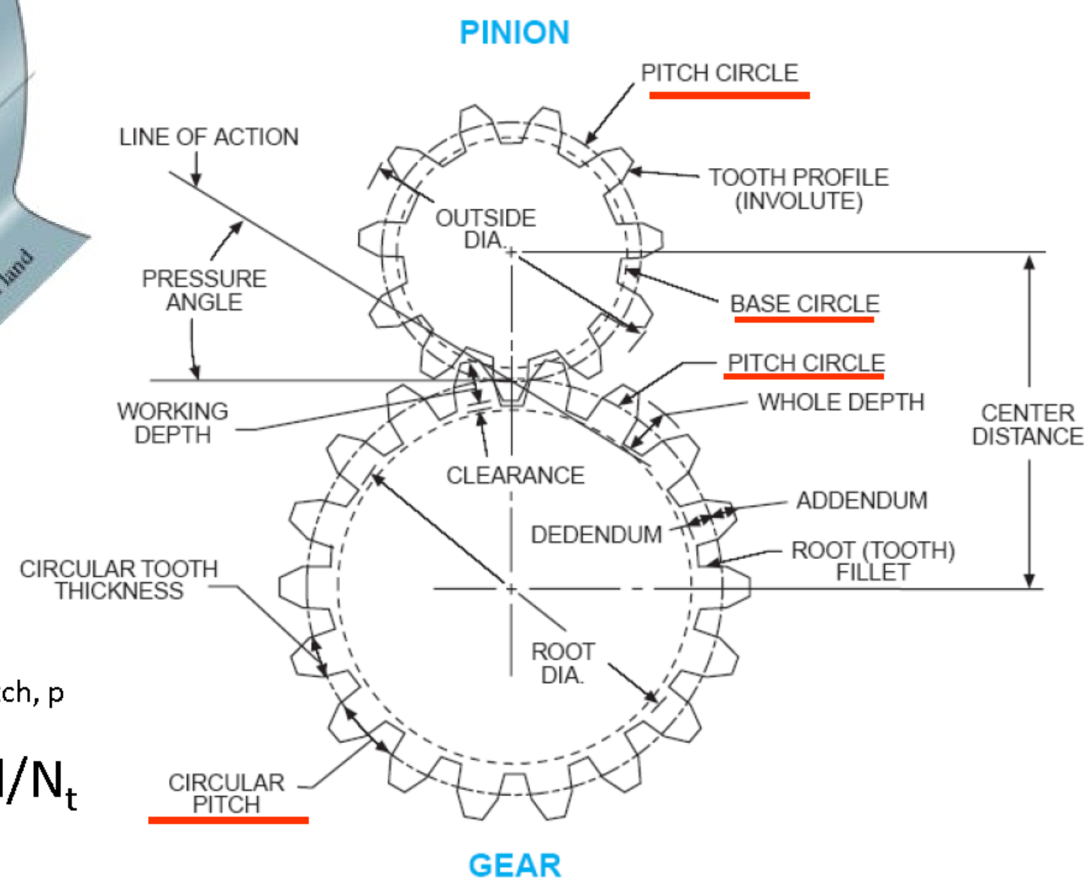


knee

Tooth Nomenclature



Driving



Driven

Circular pitch, p

$$p = \pi d/N_t$$

Nomenclature

Pitch Circle	The pitch circle is a theoretical circle upon which all calculations are usually based. Pitch circles of a pair of mating gears are tangent to each other.
Pitch Diameter, d	The diameter of the pitch circle. It depends on the spacing between gear centers and is not an intrinsic dimension of an individual gear.
Pinion	Smaller of the two mating gears
Gear	Larger of the two mating gears
Number of Teeth, N	
Circular Pitch, p	The distance, measured on the pitch circle, from a point on one tooth to a corresponding point on the adjacent tooth which is equal to the sum of the tooth thickness and the width of the space. The pitch of the two gears in the mesh must be identical.

$$\boxed{p = \pi\, d_G / N_G = \pi\, d_P / N_P \text{ and } p\,P = \pi}$$

Module, m	The ratio of the pitch diameter to the number of teeth. The customary unit of length used is the millimeter. The module is the index of tooth size in SI. $m = d / N$
Diametral pitch, P	The ratio of the number of teeth on the gear to the pitch diameter. This is the reciprocal of the module and is expressed in teeth per inch or U.S. units. $P = N / d$
Addendum, a	Radial distance between the top land and the pitch circle.
Dedendum, b	Radial distance form the bottom land to the pitch circle.
Center Distance, C	The distance from the center of the pinon to the center of the gear the sum of the pitch radii of two gears in mesh.
Backlash	The amount by which the width of a tooth space exceeds the thickness of the engaging tooth measured on the pitch circles.

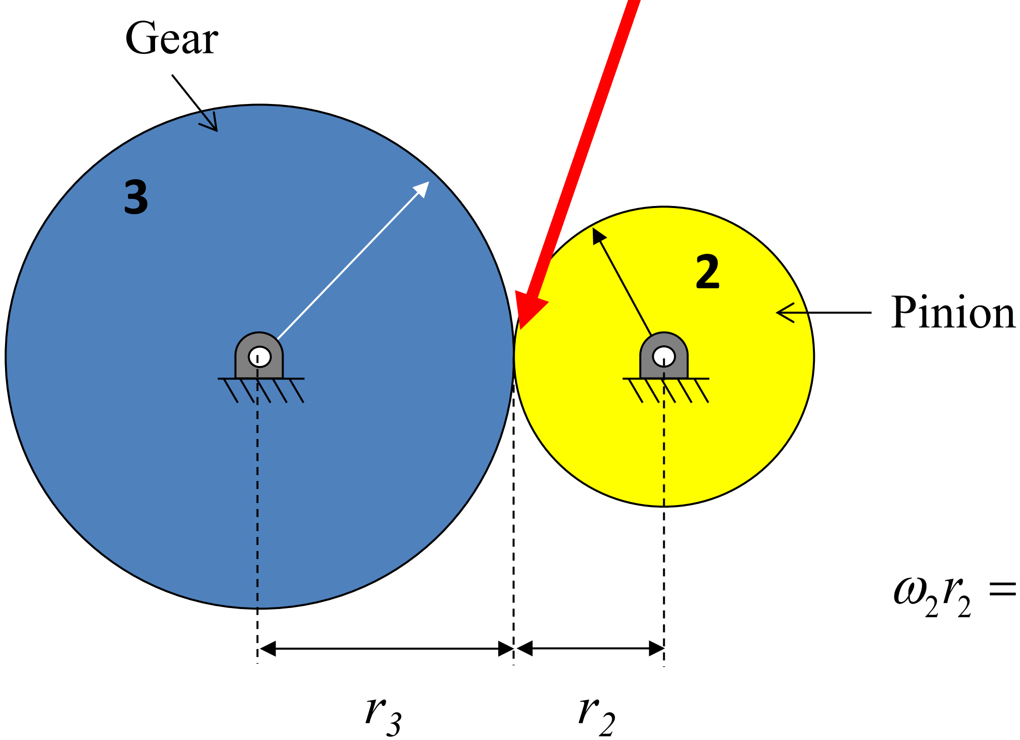
(space width of one gear) - (tooth thickness of other gear)

Nomenclature[5]

Whole depth, h_t	The radial distance from the top of a tooth to the bottom of the tooth space, $h_t = a - b$
Working depth, h_k	The radial distance that a gear tooth projects into the tooth space of the mating gear, $h_k = 2a$
Clearance circle	Circle that is tangent to the addendum circle of the mating gear.
Clearance, c	The amount by which the dedendum in a given gear exceeds the addendum of its mating gear or (dedendum of one gear) - (addendum of other gear)
Outside Diameter, d_o	The diameter of the circle that encloses the outside of the gear teeth. $d_o = d + 2a = (N + 2) / P = m N + 2m = m (N+2)$
Root Diameter, d_r	The diameter of the circle that contains the bottom of the tooth space; this circle is called the root circle. $d_r = d - 2b$
Tooth Thickness, t	The arc length, measured on the pitch circle, from one side of a tooth to the left side of the next tooth. Theoretically, the tooth space equals the tooth thickness. But for practical reasons, the tooth space is made larger.
Face Width, F	The width of the tooth measured parallel to the axis of the gear.
Fillet	The arc joining the involute tooth profile to the root of the tooth space
Face	The surface of a gear tooth from the pitch circle to the outside circle of the gear.
Flank	The surface of a gear tooth from the pitch circle to the root of the tooth space, including the fillet.
Rack	A spur gear with infinite pitch diameter. The rack does not rotate it translates.
Internal or annular gear	A gear with inside teeth can be thought of a spur gear with a negative pitch diameter ($d<0$)

Review: Fundamental Law of Gearing

Assuming roll without slip at mating point:



$$\omega_2 r_2 = \omega_3 r_3 \Rightarrow \frac{\omega_3}{\omega_2} = \frac{r_2}{r_3} = \frac{N_2}{N_3}$$

$N \propto r$

Because for mating gears

Important Definitions for Gear Trains

Pinion: Smaller of the two mating gears

Gear: Larger of the two mating gears

N: Number of teeth (Note: different from gear ratio)

Externally and Internally Mating Gears

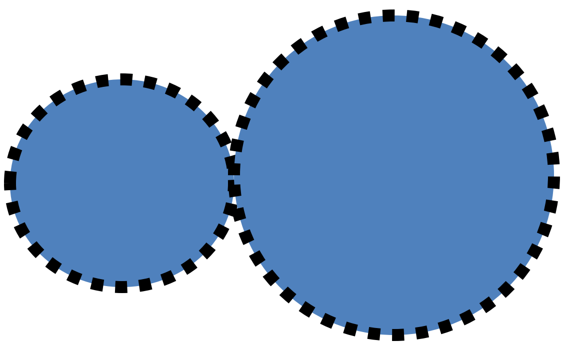
Fundamental law applies to both cases; the only difference is the direction of motion

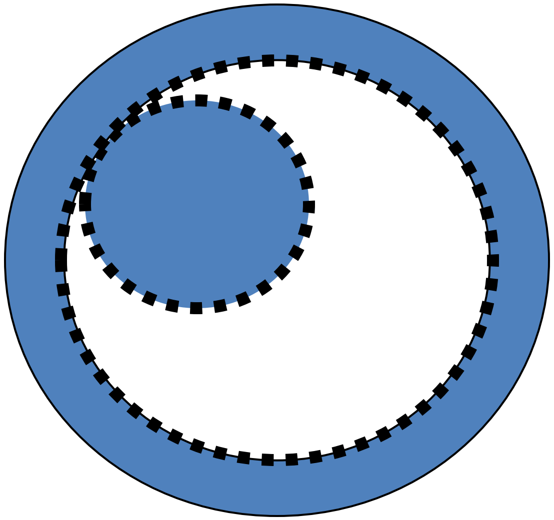
$$\omega_b = -\omega_a \frac{N_a}{N_b}$$

Externally mating:
Gears spin in opposite directions

$$\omega_b = \omega_a \frac{N_a}{N_b}$$

Internally mating:
Gears spin in same direction





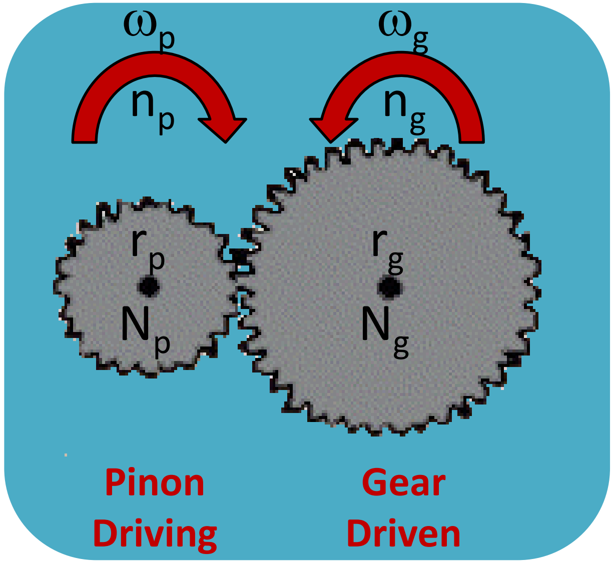
Mike Cherry

Velocity Ratio

What is important is maintain the relative ratios. Using the first definition, from the nomenclature the velocity ratio can be written in many ways:

- ω is angular speed
- r is pitch radius
- n is rpm
- d is pitch diameter
- N is number of teeth
- θ is angular position
- α is angular acceleration
- T is torque

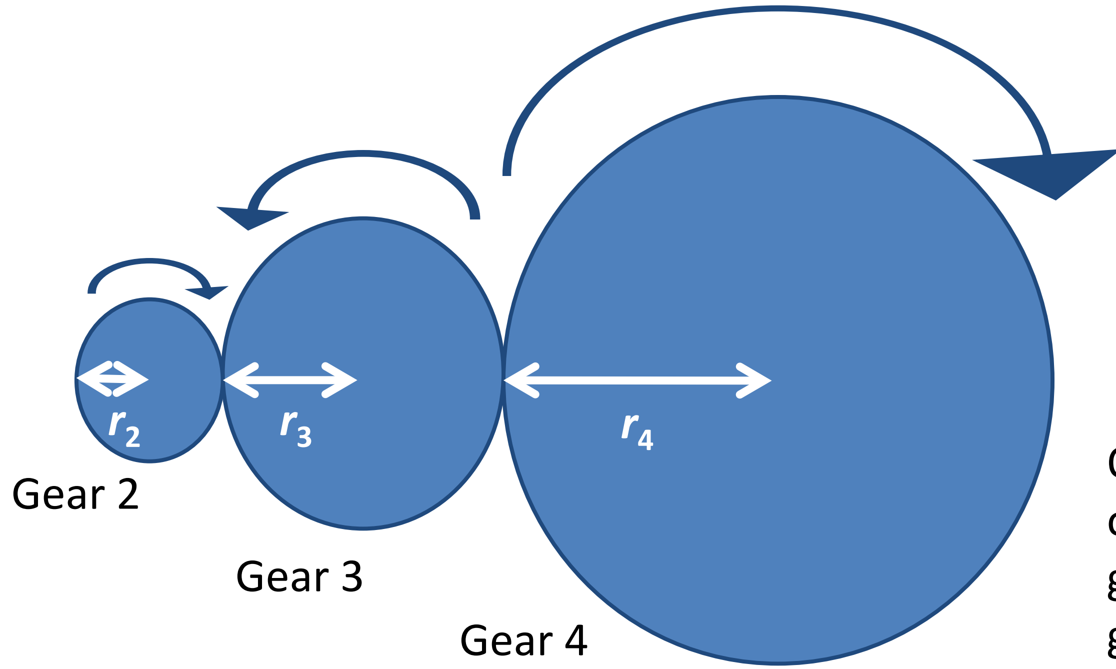
Convention is clockwise positive.



$$\omega_G / \omega_P = r_P / r_G = n_G / n_P = d_P / d_G = N_P / N_G = \theta_G / \theta_P = \alpha_G / \alpha_P = T_P / T_G$$

Most gears are speed reducers, that is the output speed is lower than the input speed.

Simple Gear Train: One Gear for each Shaft

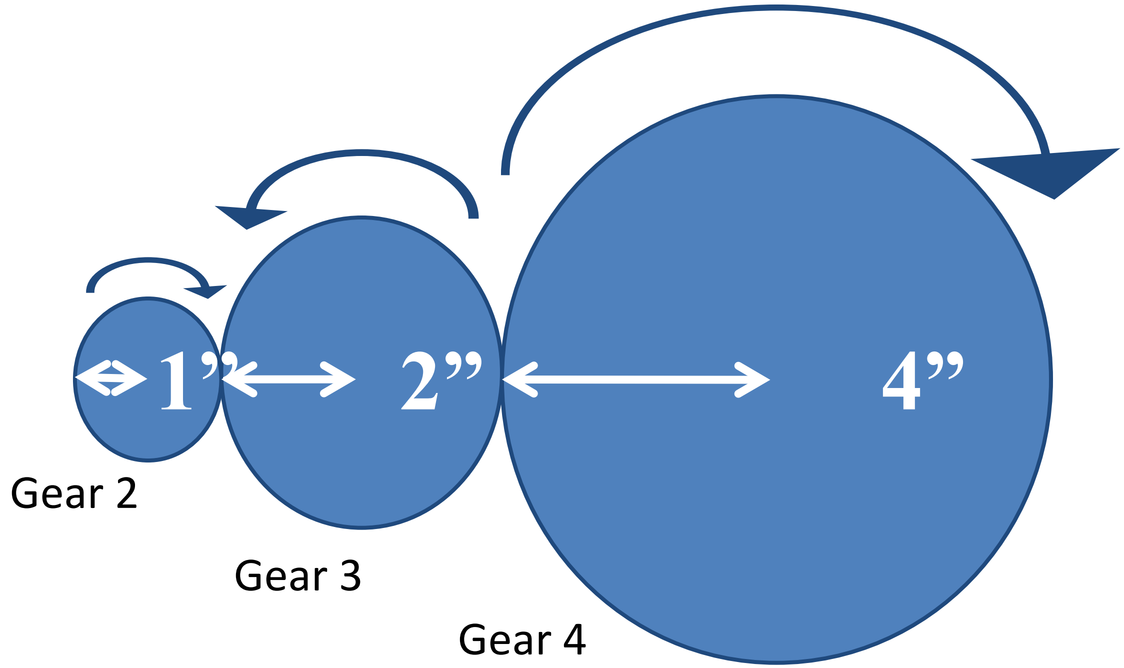


Only the input and output gears affect gear ratio in simple gear trains

$$\left|\frac{\omega_{out}}{\omega_{in}}\right| = \frac{\prod N_{driving}}{\prod N_{driven}} = \frac{N_2}{N_3}\frac{N_3}{N_4} = \frac{N_2}{N_4} = \frac{r_2}{r_4}$$

Note: Gear 3 is called the **idler**; it does not affect the gear ratio but only helps to change spin direction

Simple Gear Train (one gear for each shaft)



If the pitch is the same, then N, d and r are all proportional.

$$\left|\frac{\omega_{out}}{\omega_{in}}\right| = \frac{\prod N_{driving}}{\prod N_{driven}} = \frac{N_2}{N_3}\frac{N_3}{N_4} = \frac{N_2}{N_4} = \frac{r_2}{r_4} = \frac{1}{4}$$

Tony Walsh

Compound Gear Train

- A practical limit on train value for one pair of gears is 10 to 1
- To obtain more, can "compound" two gears onto the same shaft.

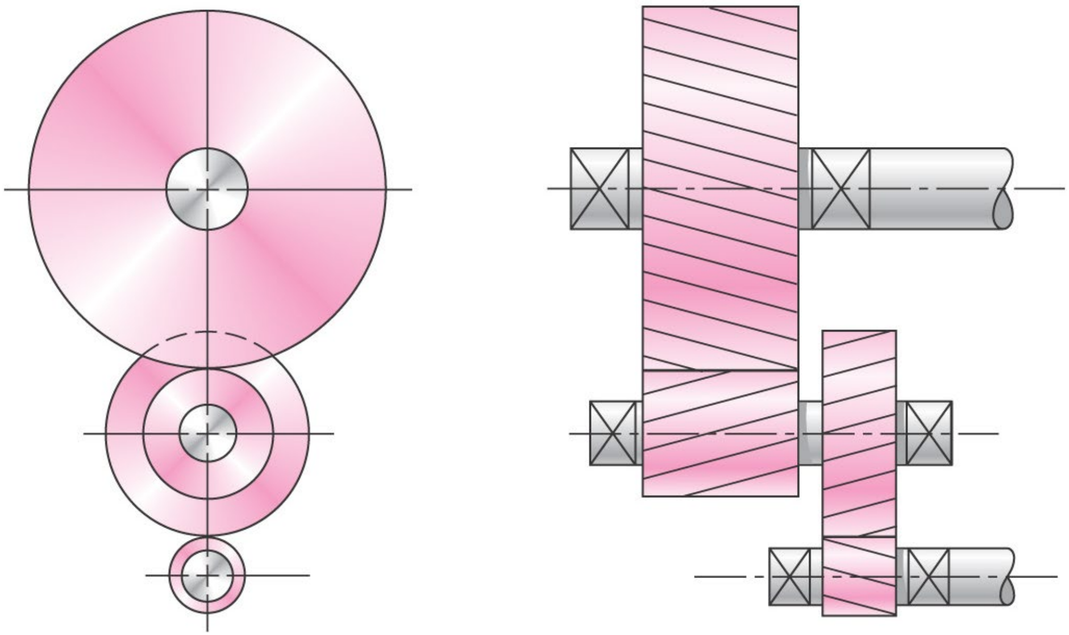
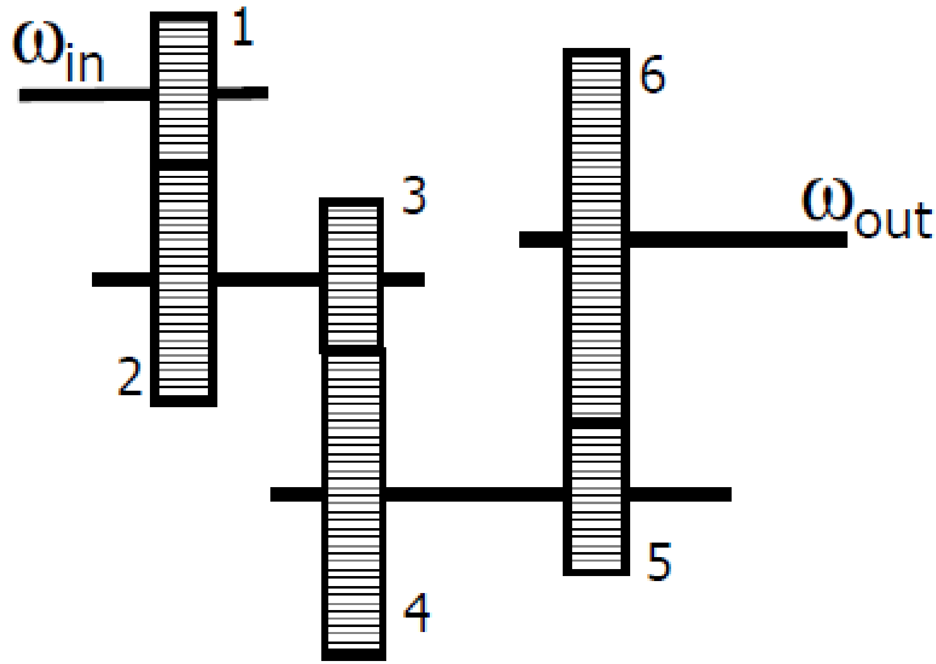


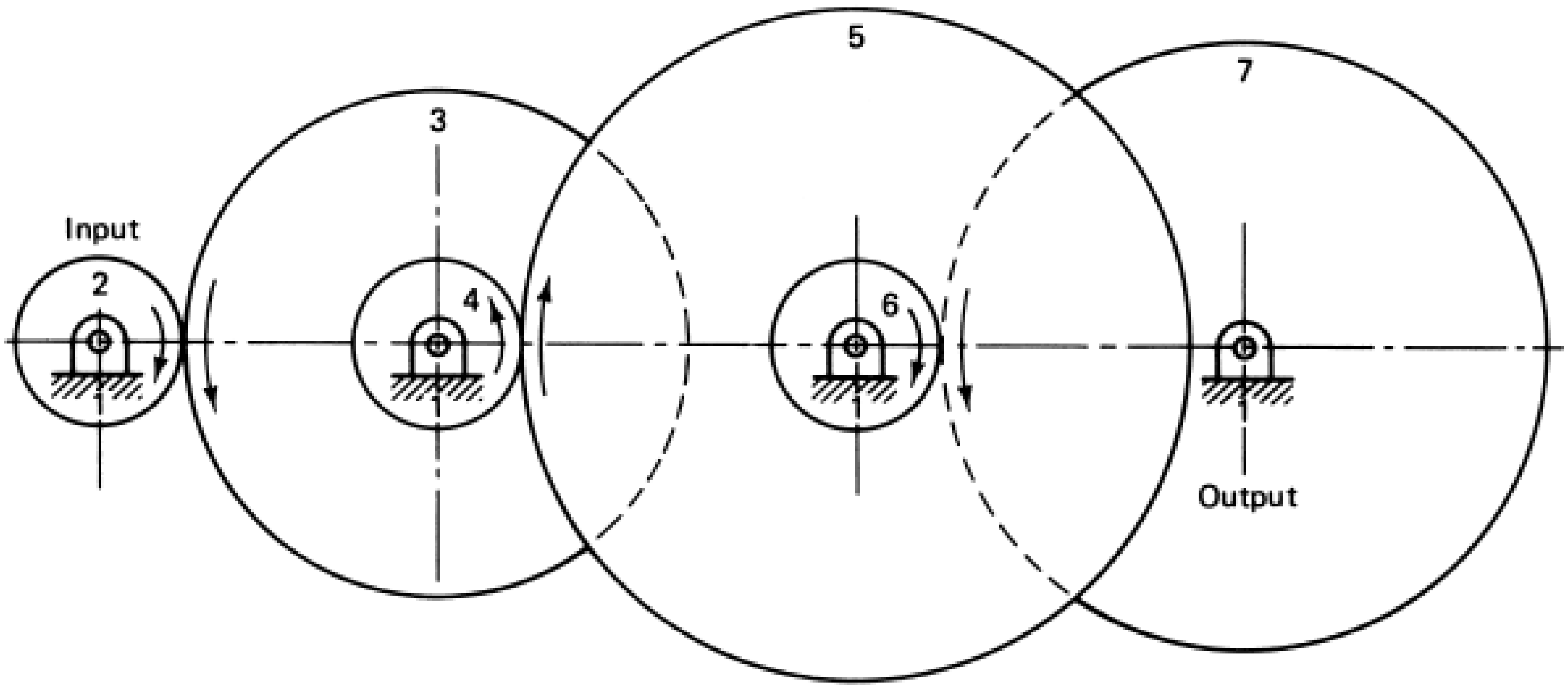
Fig. 13–28

Compound Gear Trains

A compound gear train has two or more gears rigidly attached to the same shaft.



Compound Gear Trains

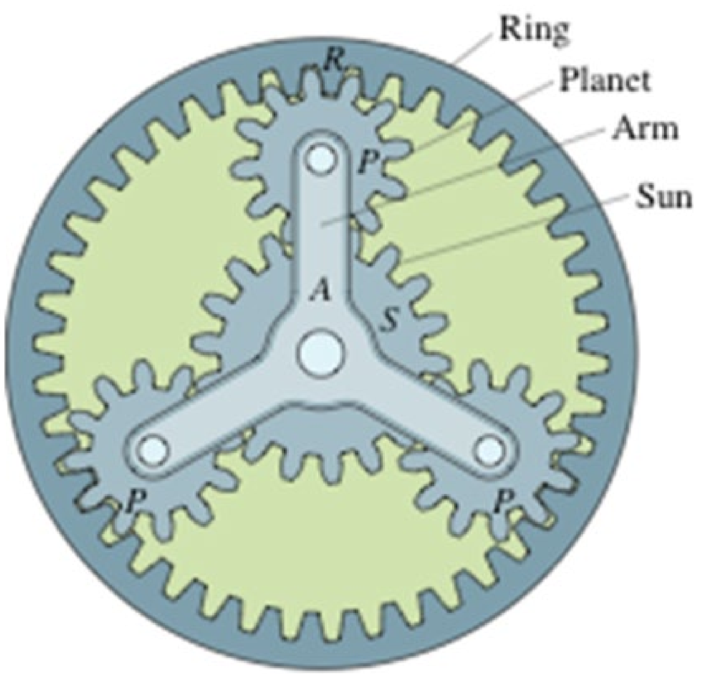


$$\left|\frac{\omega_{out}}{\omega_{in}}\right| = \frac{\prod N_{driving}}{\prod N_{driven}} = \frac{N_2}{N_3}\frac{N_4}{N_5}\frac{N_6}{N_7}$$

Direction can be found by inspection.

Planetary Gear Train

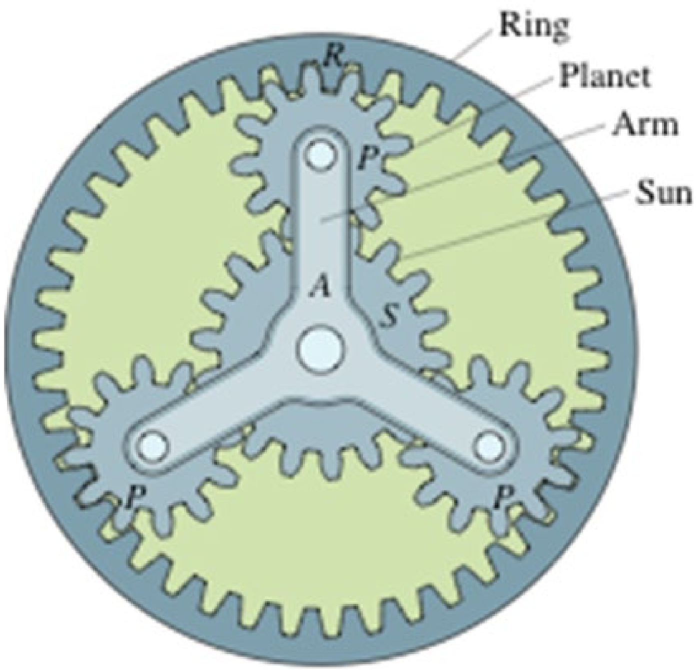
- A planetary gear train consists of four main members
  - sun gear
  - ring gear
  - planetary carrier (arm)
  - planetary gears



- The planetary carrier holds gears in proper relation with the sun and ring gear.
- The planetary gears are free to rotate on their own axis while they "walk" around the sun gear or inside the ring gear.

ME 350 Puzzle

In which of the following transmissions is the planetary gear used?



A) Automatic transmission in cars

B) Manual transmission in cars

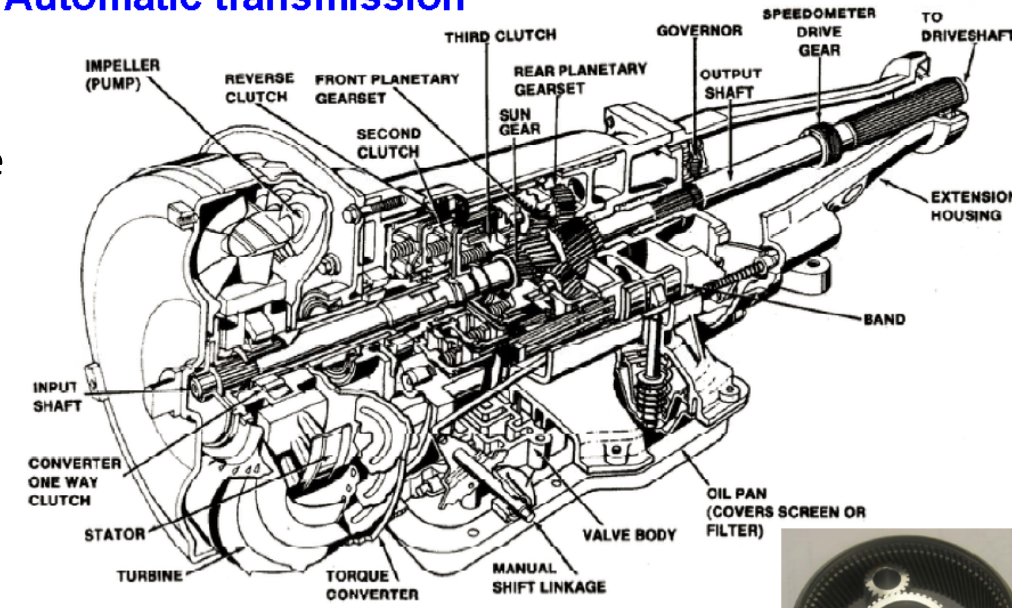
C) Transmission in ME 350 motors

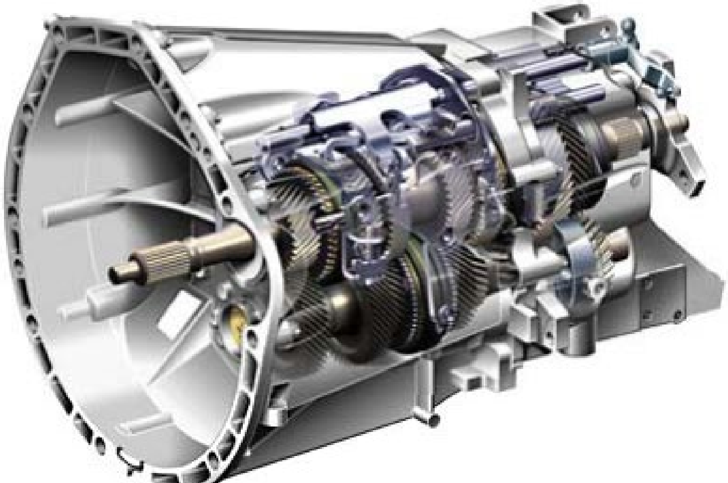
D) Not sure

Car Transmissions

Automatic transmission

Planetary gear sets are a hallmark of automatic transmissions, where they allow for smooth, electronically controlled gear changes through internal clutch packs



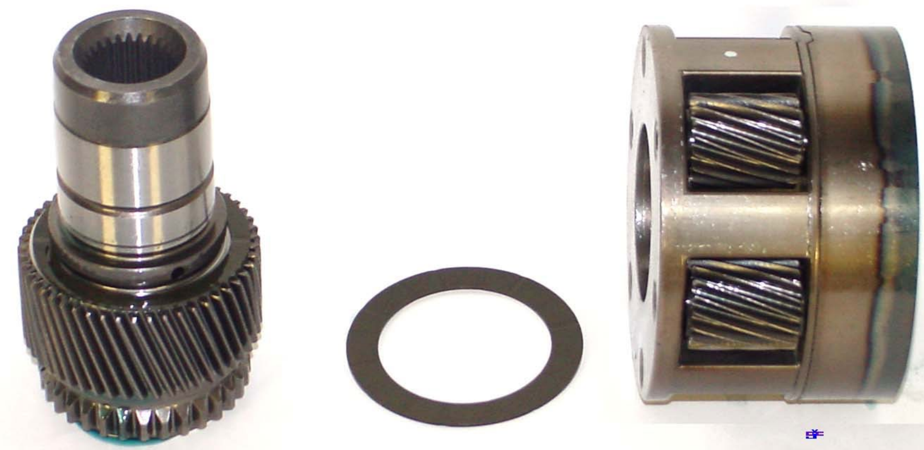


Manual Transmission

a traditional manual car transmission does not use a planetary gear set; instead, it uses parallel-axis spur gears and a clutch system to change gears. Manual transmissions rely on a driver-operated clutch pedal and a physical shifter to directly engage specific gear pairs.

Planetary Gears

- By holding or releasing the components of a planetary gearset, it is possible to do the following:
  - Reduce output speed and increase torque (gear reduction).
  - Increase output speed while reducing torque (overdrive).
  - Reverse output direction (reverse gear).
  - Serve as a solid unit to transfer power (one to one ratio).
  - Freewheel to stop power flow (park or neutral).

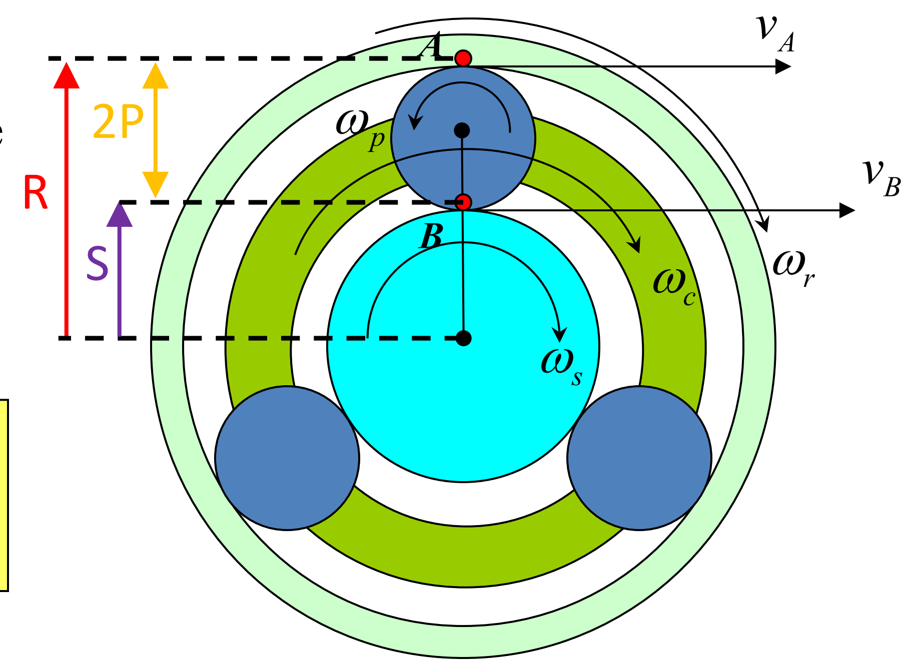


http://auto.howstuffworks.com/automatic-transmission3.htm

Kinematics of Planetary Gear Train

Notice that ω_p is defined positive in the CCW direction while other angular velocities are positive in CW. This sign convention must be used in the equation below

$$\boxed{\omega_c = \omega_s \frac{S}{S+R} + \omega_r \frac{R}{S+R}}$$



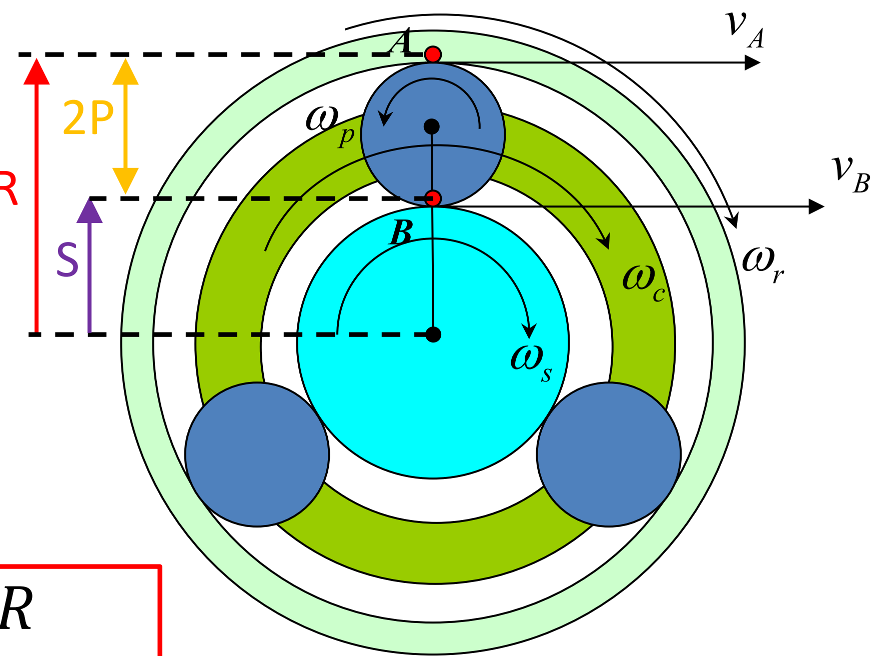
S, P & R = # of teeth (or radius) on the Sun, Planets and Ring Gears

Kinematics of Planetary Gear Train

We know from <u>ME 240</u>:

$$(1)\quad v_A = \underbrace{\omega_r \cdot R}_{\text{On ring}} = \underbrace{\omega_c(S+P) - \omega_p P}_{\text{On planet}}$$

$$(2)\quad v_B = \underbrace{\omega_s \cdot S}_{\text{On Sun}} = \underbrace{\omega_c(S+P) + \omega_p P}_{\text{On planet}}$$



<u>Adding (1) and (2):</u>

$$(3)\quad \omega_r \cdot R + \omega_s S = 2\omega_c\,(S+P)$$

<u>Geometrical relationship:</u>

$$R = 2P + S \qquad (4)$$

<u>Put Eqn. (4) in (3) and simplify:</u>

$$\boxed{\omega_c = \omega_s \frac{S}{(S+R)} + \omega_r \frac{R}{(S+R)}}$$

Exercise 4

The sun gear is driven clockwise at 100 rpm. The ring gear is held stationary by being fastened to the frame. Find the rpm and direction of rotation of gear 4 (the planet).

Knowns:

$S = 20\ T$

$R = 80\ T$

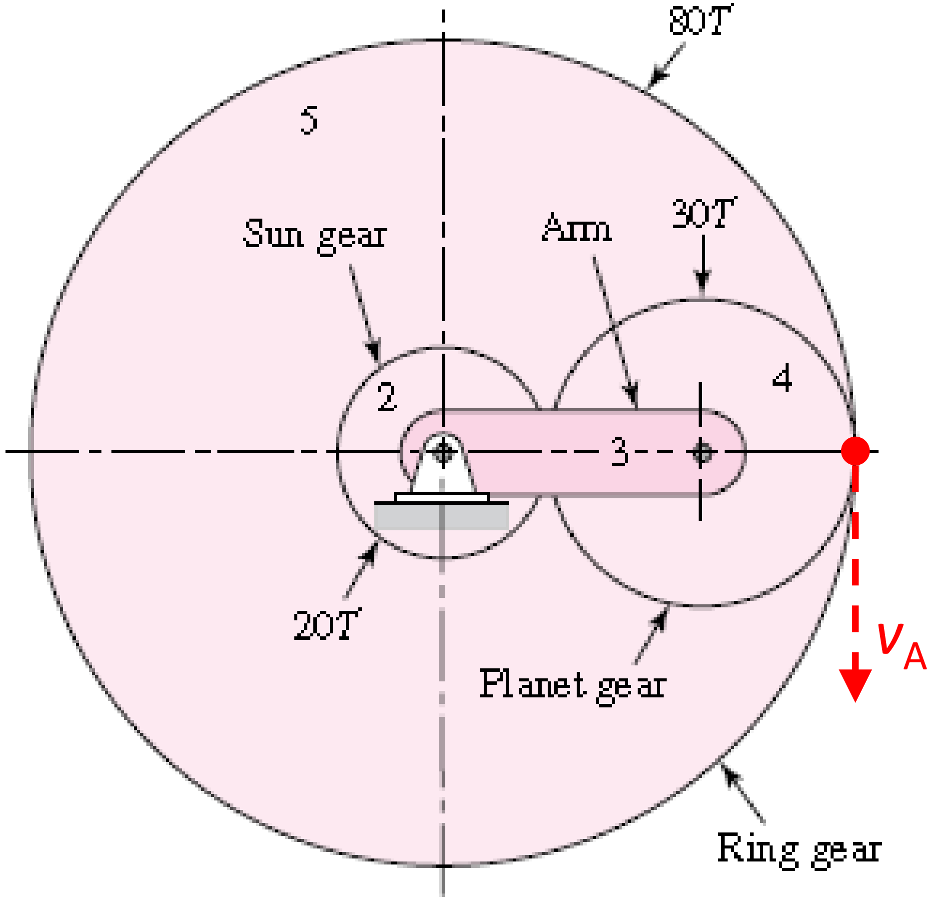
$P = 30\ T$

$\omega_r = 0$

$\omega_S = 100$ rpm

Find:

ω_p & direction

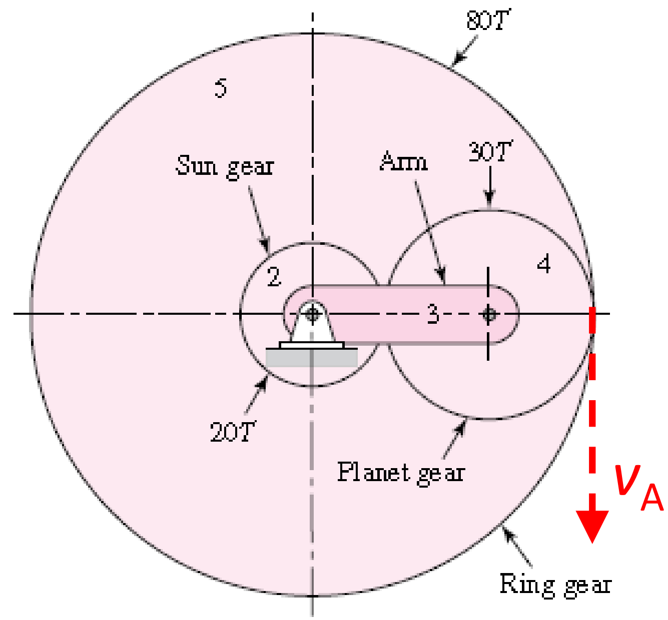


Exercise 4

$$\omega_c = \omega_s \frac{S}{(S+R)} + \cancelto{0}{\omega_r} \frac{R}{(S+R)}$$

$$\rightarrow \omega_c = \omega_s \frac{S}{(S+R)} = 100\ rpm \cdot \frac{20}{(20+80)}$$ = **20 rpm**

Recall: at point A (on carrier and planet)



$$v_A = \underbrace{\omega_r \cdot R}_{\text{On ring}} = \underbrace{\omega_c (S+P) - \omega_p P}_{\text{On planet}}$$

$$\cancelto{0}{\omega_r} \cdot R = \omega_c (S+P) - \omega_p P$$

$$\therefore \omega_p = \frac{\omega_c (S+P)}{P} = 20\ rpm \cdot \frac{(20+30)}{(30)}$$ = **+33.33 rpm (CCW +)**